7 Finance

7.4 Loans

Formulas so far:

For these formulas, we have the following: P_0 is the principal. r is the interest rate. t is the number of time intervals (specified by problem) which have passed. A is the amount paid back. I is the interest.

Simple Interest	$A = P_0(1+r)$
Simple Interest over Time	$A = P_0(1+rt)$

For these formulas, we have the following: P_0 is the principal. r is the annual interest rate. t is the number of years which have passed. A is the total amount. k is the number of times interest is compounded per year. d is the amount we deposit (or withdraw) from the account regularly.

Compound Interest	$A = P_0\left(1+\frac{r}{k}\right)^{kt}$
Savings Annuity	$A = d\frac{\left(\left(1+\frac{r}{k}\right)^{kt}-1\right)}{\frac{r}{k}}$
Payout Annuity	$P_0 = d\frac{\left(1-\left(1+\frac{r}{k}\right)^{-kt}\right)}{\frac{r}{k}}$

Example 7.4.1 Suppose that I take out an auto loan from the bank. What does the formula for this situation look like?

Example 7.4.2 You want to take out a \$140,000 mortgage (home loan). The interest rate on the loan is 6%, and the loan is for 30 years. How much will your monthly payments be?

Example 7.4.3 Today's mortgage rates are around 6.50%. Suppose you take out a \$250,000 mortgage. What will your monthly payment be to the bank? What is the total amount you will pay the bank over the next 30 years?

Example 7.4.4 Mortgage rates tend to be between 6.13% and 7.26%, and is affected by many things such as the borrower's credit score, the amount of the loan, inflation, etc. Suppose that you wish to take out a mortgage of \$300,000 from the bank of your choice. Compare the outcomes of the following two scenarios:

1. You take out a 30 year loan, with payments made monthly, during a year in which rates are slightly higher, at 6.85%. What is the total amount you will pay the bank over the next 30 years?
2. You wait a year, and the rate you are offered now by the bank is 6.43%. The loan is still 30 years with payments made monthly. What is the total amount you will pay the bank over the next 30 years?

For the first, $d_1 = 1965.78$, and the second $d_2 = 1882.41$. Total amounts paid are $A_1 = 707680.80$ and $A_2 = 677667.60$. That's a \$30013.20 difference!

Example 7.4.5 Suppose that you have a 5 year auto loan for which you pay \$325 per month. If the interest rate is 5%, how much will you still owe the bank after 3 years of making payments? (It helps to think of this one from this perspective: If the bank invests money into you, and they wish to make \$325 'withdrawals' once a month for 2 years, how much do they need to have in the account).

Example 7.4.6 A couple purchases a home with a \$180,000 mortgage at 4% for 30 years with monthly payments. What will the remaining balance on their mortgage be after 5 years? (hint: you first need to know what their monthly payment is, then it's just like the previous example)